Faculty of Applied Science and Technology

LAB #2 – Array Operations in MATLAB

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Procedures and results:

Add the number of all questions in this report, even if you decided to not answer them.

For each question, you should explain your observation + code snippet.

1. With MATLAB, do the problems 4 to 8 in your textbook chapter 2 (section 2.11) and problems 3 to 5 of chapter 3 (section 3.9).

Q4) Create the variable d that is a column vector with the following elements:

d = [0.75*5.2^0.7; 11.1; nthroot(60,3); tan((10*pi)/11); cosd(5)^2; 0.116];

Q5) Defined the variables x= 3.4 and y=5.8, and then used them to create a row vector (assign it to a variable named e) that has the following elements: x/y, x+y, x^y, x*y, y^2, and x.

x = 3.4; y = 5.8; e = [x/y; x+y; x^y; x*y; y^2; x];

Q6) Defined the variables c=4.5 and d=2.8, and then used them to create a row vector (assign it to a variable named f) that has the following elements: d^2, c, (c+d), c^d and d.

c = 4.5; d = 2.8; e = [d^2; c; (c+d); c^d; d];

Q7) Created a variable g that is a row vector in which the first element is 3 and the last element is 27, with an increment of 4 between the elements (3, 7, 11, … , 27).

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g = [3:4:27];
```

Q8) Created a variable h that is a row vector with eight equally spaced elements in which the first element is 68 and the last element is 12.

```
h = linspace(68,12,8);
```

Q3) Calculated the value of y for the following values of x using element-by-element operations: –2, –1.5, –1, –0.5, 0, 0.5, 1, 1.5, 2.

$$y = (x + x\sqrt{x+3})(1 + 2x^2) - x^3$$

```
x = [-2:0.5:2]
y = (x+nthroot(x+3,x)).*(1+2.*x.^2)-x.^3
```

```
y = -1.0000   -0.6777   -0.5000   -0.3850   Inf   19.0000   14.0000   19.8662   30.1246
```

Q4) Calculated the value of y for the following values of x using element-by-element operations: 15, 25, 35, 45, 55, 65.

```
x = [15:10:65];
y= (4.*sind(x+6))./((cosd(x).^2+sind(x)).^2)
```

```
y = 1.0092    1.3312    1.6942    2.1334    2.6539    3.2132
```

Q5) Determined the radius and surface area of spheres with volumes of 4,000, 3,500, 3,000, 2,500, 2,000, 1,500, and 1,000.

```
V = [4000:-500:1000];
r = round(nthroot( ((3.*V)./(4.*pi)),3), 1)
S = round((4.*pi.*r.^2), 1)
disp([r;V;S]')
```

```
r = 9.8000    9.4000    8.9000    8.4000    7.8000    7.1000    6.2000
```

S = 1.0e+03 *
    1.2069   1.1104   0.9954   0.8867   0.7645   0.6335   0.4831

1.0e+03 *
    0.0098   4.0000   1.2069
    0.0094   3.5000   1.1104
    0.0089   3.0000   0.9954
    0.0084   2.5000   0.8867
    0.0078   2.0000   0.7645
    0.0071   1.5000   0.6335
    0.0062   1.0000   0.4831

2. In your textbook, look at the more advanced problems after number 20 (section 2.11-chapter 2), select 22, 23, 24, 37 and 38 from those and do them with MATLAB.

Q22) Created two row vectors vD=20:4:44 and vE=50:3:71. Then, create the following new vectors by assigning elements of vD and vE to the new vec_xfffe_tors:

vD=20:4:44;

vE=50:3:71;

A) A vector (name it vDE) that contains the 2nd through the 5th elements of vD and the 4th through 7th elements of vE;

vDE = [vD(:,2:5) vE(:,4:7)]

vDE = 24   28   32   36   59   62   65   68

B) A vector (name it vED) that contains elements 6, 5, 4, 3, and 2 of vE and elements 4, 3, 2, and 1 of vD;

vED = [vE(:,6:-1:2) vD(:,4:-1:2)]

vED = 65   62   59   56   53   32   28   24

Q23) Created a nine-element row vector vF=5:7:61. Then create a vector (name it vFrev) that consist of the elements of vF in reverse order.

vF = 5:7:61
vFrev = vF(end:-1:1)

vF = 5   12   19   26   33   40   47   54   61

vFrev = 61   54   47   40   33   26   19   12   5

Q24) Create the following matrix by assigning vectors with constant spacing to the rows (use the linspace command for the third row).

A = [1:1:7; 7:-1:1; linspace(2,9,7)]

A =

1.0000	2.0000	3.0000	4.0000	5.0000	6.0000	7.0000
7.0000	6.0000	5.0000	4.0000	3.0000	2.0000	1.0000
2.0000	3.1667	4.3333	5.5000	6.6667	7.8333	9.0000

Q37) Created the following matrix M:

M = reshape(1:2:29,3,5)

M =

1 7 13 19 25

3 9 15 21 27

5 11 17 23 29

A) Created a five-element row vector named Va that contains the elements of the third row of M.

Va = M(3,:)

Va = 5   11   17   23   29

B) Created a three-element column vector named Vb that contains the ele_xfffe_ments of the fourth column of M.

Vb = M(:,4)' % I put the asterix for Transform to just make it easier to read can be removed.

Vb = 19   21   23

C) Created an eight-element row vector named Vc that contains the ele_xfffe_ments of the second row of M followed by the elements of the third col_xfffe_umn of M

Vc = [M(2,:), M(:,3)']

Vc = 3   9   15   21   27   13   15   17

Q38) Created the following matrix N:

N = reshape(0:3:51,6,3)'

N =

0	3	6	9	12	15
18	21	24	27	30	33
36	39	42	45	48	51

A) Created a six-element row vector named Ua that contains the first three elements of the first row of N followed by the last three elements of the third row of N.

Ua = [N(1,1:3) N(3,4:6)]

Ua = 0 3 6 45 48 51

B) Created a nine-element column vector named Ub that contains the ele_xfffe_ments of the first column of N, followed by the elements of the third col_xfffe_umn of N, followed by the elements of the sixth column of N.

Ub = [N(:,1)' N(:,3)' N(:,6)']

Ub = 0 18 36 6 24 42 15 33 51

C) Created a six-element column vector named Uc that contains elements 2, 3, 4, and 5 of the second row of N, followed by elements 2 and 3 of the fifth column of N.

Uc = [N(2,2:5) N(2:3,5)']

Uc = 21 24 27 30 30 48

4. Tracing Exercise:

Evaluate what will be the values of p, q, and r after these MATLAB instructions will be executed (Do these by hand: Do not use MATLAB!). Must show all your calculation procedures.

n1 = [1:5:9];	n2 = [10 14 4];	n3 = [length(n1) 0 0];
[1, 6]	[10, 14, 4]	[2, 0, 0]

p = [n1*3, n2, n3-n2]; P = [3, 18, 10, 14, 4, -8, -14, -4]

n4 = [10, 16, 3.3, 6];	m1 = [1 2 3 4; 5 6 7 8];	m2 = [m1;n4];
[10, 16, 3.3, 6]	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8 10 16 3.3 6

q = ones (4,3) + m2';

q =	2	6	11
	3	7	17
	4	8	4.3
	5	9	7

n5 = [10:20:60];	m3 = [eye(3) .* 10; n5];	s = [m3(3,2:3), m3(1:2,3)'];	s(3) = 12;
[10, 30, 50]	10 0 0 0 10 0 0 0 10 10 30 50	[0, 10, 0, 0]	[0, 10, 12, 0]

r = [m3(3,:), s]; r = [0, 0, 10, 0, 10, 12, 0]

5. Finished the required steps

References:

MATLAB LOGIN: Matlab & Simulink. MATLAB Login | MATLAB & Simulink. (n.d.). https://matlab.mathworks.com/

Gilat, A. (2017). MATLAB: An introduction with applications (6th ed.). John Wiley & Sons.